Topological characteristics of urban road networks - A case study in São Paulo, Brazil

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1 Introduction

Road networks are vulnerable to disasters, such as floods, which can cause damage to infrastructure and adversely affect travel and/or local populations on degraded networks. Studying and analyzing the topological characteristics of road networks helps prioritize planning, budgeting, and maintenance, as well as prepare effective emergency response plans [1]. Each city has its own characteristics of road design, intersection density, mobility patterns, and population distribution, which can directly influence results.

In this scenario, the objective of this study is to assess the topological characteristics of urban road network. Using the city of São Paulo, Brazil, as a case study, this manuscript explores topological properties in the studied area.

Previous studies have investigated the vulnerability of transport and road networks using graph-based approaches and the analysis of flood occurrence. Regarding road network topology, previous studies have focused on vulnerability in urban road networks [4] and rural road networks [5], analyzing vulnerability to flooding in mobility systems.

2 Material and methods

2.1 Study area

2.2 Graph analysis

The road network presented in this work is represented by a graph G, where each intersection is a node and each road segment is an edge, therefore, V is a set of vertices (or nodes) and E is a set of edges (or links) that connects pairs of nodes (or connects a node to itself): $G = (V, E)$. The edge (e) betweenness centrality ($c_B(e)$) is calculated by the sum of the fraction of all-pairs shortest paths that pass through that edge, as the following Equation shows:

$$c_B(e) = \sum_{s,t \in V} \frac{\sigma(s,t|e)}{\sigma(s,t)}; \tag{1}$$

where V is the set of nodes, $\sigma(s,t)$ is the number of shortest (s,t)-paths, and $\sigma(s,t|e)$ is the number of those paths passing through edge e [3].

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Also, the node degree (k_i) was calculated, which is the number of edges connected to that node. This is known as adjacency (a_{ij}). When two nodes i and j are said to be neighbors, then $a_{ij} \neq 0$. The following Equation 2 shows the computation of k_i [3]:

$$k_i = \sum_j a_{ij} \tag{2}$$

The clustering coefficient (c_i) of a node i quantifies the proportion of existing connections among its neighbors relative to the maximum possible number of such connections [3]:

$$c_i = \frac{2L_i}{k_i(k_i - 1)} \tag{3}$$

where L_i is the number of edges between the neighbors of node i. The average shortest path length (l_i) of a node i is the mean of the shortest distances from i to all other reachable nodes [3]:

$$l_i = \frac{1}{N-1} \sum_{j \neq i} d_{ij} \tag{4}$$

where d_{ij} is the shortest path length between nodes i and j, and N is the total number of nodes.

Average values of degree ($\langle k \rangle$), clustering coefficient ($\langle c \rangle$) and average shortest path length ($\langle l \rangle$) were calculated and compared to a random network following the $G(N, p)$ model, where N is the quantity of nodes and p is the probability that these nodes are connected [2].

3 Results

4 Conclusion

References

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